One Life of the Genome

What is in the genome, in short. How the experiment runs, in plain steps. And his whole life drawn once, from the two inputs to the verdict.

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1 · What is in the genome — 21 lines of text

The genome is a text file he can read and change. The **machinery** is fixed code that reads it and runs it; the genome itself never changes during a life. A gene the machinery does not know stops the organism from starting. Six families:

GENE	IN PLAIN WORDS
SEED	
`SEED 20260919`	The one random number. Same seed, same life.
BODY — what senses and areas exist	
`SENSE eye 24x24 grey`	A picture is 576 grey numbers.
`RETINA centre pool3 unit`	A fixed front, not learned: centre the ink, average 3×3 blocks → 64 numbers, unit length.
`SENSE ear letters 8`	A word is up to 8 letters; one slot per letter → 216 numbers.
`AREA sight · AREA words`	Two areas of cells, one per sense. Cells are born here.
`MEET sight words`	The two areas may connect. In rung 0 the meeting is edges only.
GROW — when a cell is born	
`BIRTH novelty cos<THETA_N`	Nothing in the area looks like this input → a new cell is born holding it. THETA_N is measured from the world (gate G-a), never guessed.
`BIRTH error misfires 3`	A cell keeps answering wrong three times → a new cell is born beside it.
`BUDGET 400 per arrival`	Births allowed per new set of things. A refused birth is counted, never hidden. A cell is never replaced.
`NMAX 2000`	The ceiling on cells.
WIRE — how cells connect	
`KWTA 5`	Five winners per area per moment.
`HEBB window 1`	Cells active in the same moment across the two areas get an edge.

`DELTA lr 0.05`	An edge grows when the pairing is right, shrinks when wrong.
`SHRINK — what is removed: the wrong and the redundant, never the merely unused`	
`PRUNE w<0.02 after 3 chances`	A weak edge gets three chances, then goes.
`MERGE cos>THETA_M same-answer`	Two cells that look alike and answer alike become one.
`SCALE every 1000 moments`	Synaptic scaling: total strength kept in bounds.
`(no disuse decay)`	An unused cell or edge is never removed for being unused. That was Alpha 3's forgetting.
`STAGE — development in order`	
`STAGE 1 senses 0..200`	First 200 moments: the areas grow; no wiring between them yet.
`STAGE 2 meet 200..`	From moment 200: wiring between the areas is allowed.
`ANSWER — how a question is answered from what grew`	
`RECALL k=1`	Light one side; read the strongest edge to the other side.
`ABSTAIN second/best > 0.85`	Two answers too close to call → «I do not know this».

> Fixed and never learned: the retina, the ear, the machinery. Grown by the stream alone: every cell, every edge. The seat writes none of them.

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2 · The instructions — how the experiment runs

1. **Write the genome** as the text file above. The machinery reads it. An unknown gene → it will not start.
2. **Make the world.** Ten shapes (circle, square, triangle, cross, ring, bar, dot, star, arrow, wave), each with its name. Draw each shape 60 times with small jitter (position, size, tilt, noise). A **moment** is one picture and its word shown together.
3. **Gate G-a — measure before choosing.** Pass every picture through the retina. Measure how alike two pictures of the same thing are, and how alike pictures of different things are. Set THETA_N between those two bands, THETA_M above the first. If the bands overlap, stop and report: the retina is too weak (the fix is a stronger fixed front, never a learned one).
4. **The toy first.** A two-thing world (circle, bar) must answer Q1 perfectly. If it does not, the machinery is wrong, not the genome.
5. **The life.** 600 moments in shuffled order. Moments 0–200: the areas grow. Moments 200–600: the areas wire together. Nobody writes a cell; the stream does.

6. **Arrival 2.** Five new things (moon, house, tree, cup, key) for 300 more moments, under a fresh budget. He must keep the first ten.
7. **Ask five questions.** Q1 what is this (a new picture) · Q2 show me a «word» · Q3 is this a «word»? · Q4 which of these two is a «word»? · Q5 a thing never shown → he must say «I do not know».
8. **Score outside him.** A ledger the organism never reads knows which thing each cell was born on. He returns a cell; the ledger says whether it was right.
9. **Run the twins on the same road,** each blind to one thing: words paired at random, growth frozen, wiring off, twenty random genomes, one planted made-up pair («zog»), one pair never shown.
10. **Read the nine marks.** All must pass. The bars are written now, before any number is seen. A failed bar is a finding, never moved.
11. **Replay.** The same genome and the same stream, run twice → the same organism, byte for byte.
12. **If it passes:** rung 1 is his test — he drops a few pictures per word into `/data/dev/genome/world/<word>/` and asks through a small door (`ask what-is picture.png`, `ask show cat`). Rung 2, only after that: a lineage — mutate the genome's numbers, select on held-out families, the random-genome control kept.

> **What "it works" means, in one sentence.** Grown from the genome text alone, he names things he met with picture and word together at 0.80 or better on pictures he never saw, says «I do not know» on things never shown, and a second life from the same genome and stream is byte-identical.

3 · His life, from inputs to outputs

Time runs left to right. The top row is the road; the middle row is what he holds at each step; the dashed row is the seven control runs beside him; the last row is the nine marks the verdict is read from.

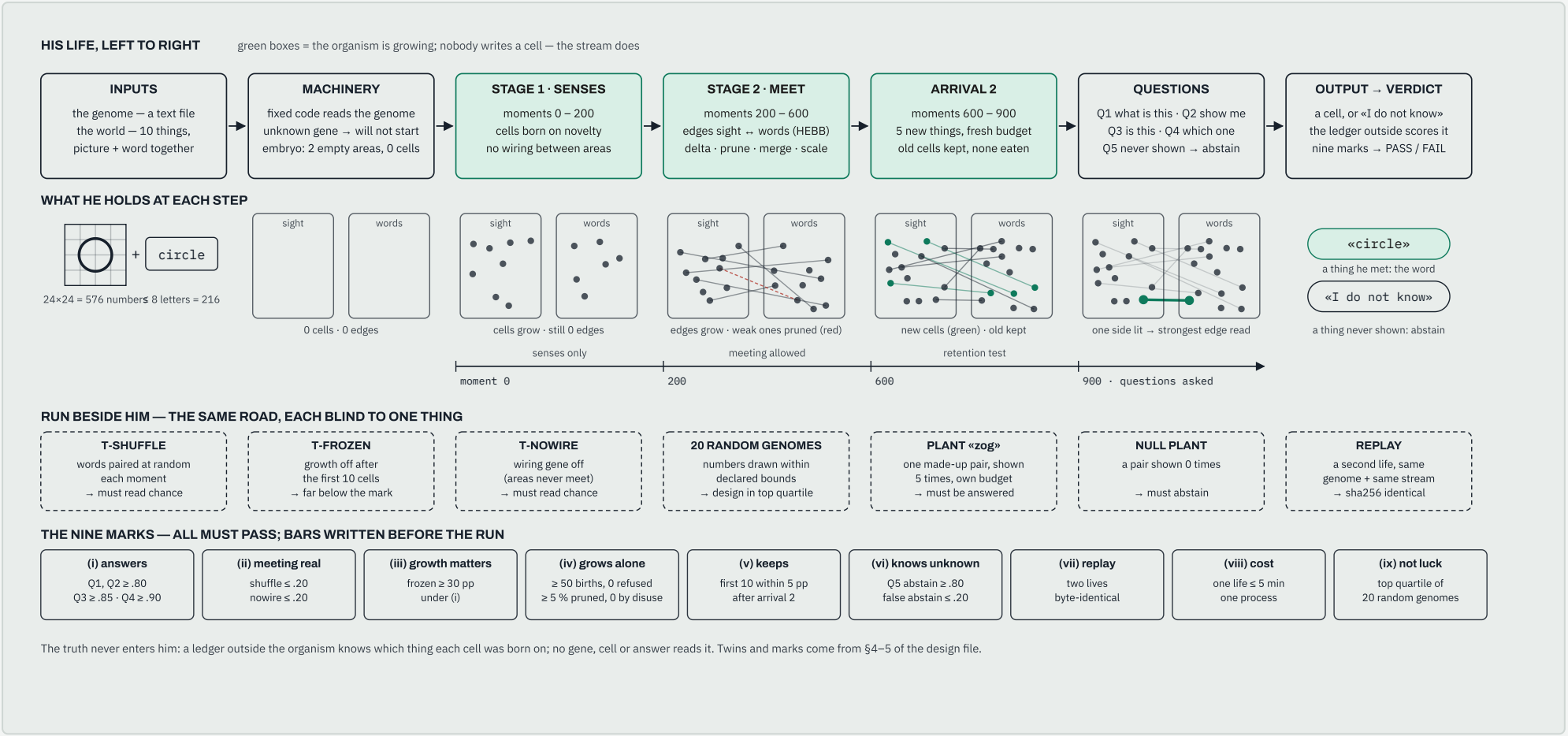


Figure 1 — One life of the genome organism, from the two inputs (a genome file, a world of pictures with words) to the verdict. Nothing between the machinery and the questions is written by the seat: cells and edges come from the stream, under the genome's rules.

4 · Inside one moment

This is the loop that runs 900 times in one life. The picture and the word arrive together; each passes a fixed front into its own area; each area either births a cell or picks five winners; from moment 200 the winners of the two areas are wired; housekeeping removes the wrong and the redundant.

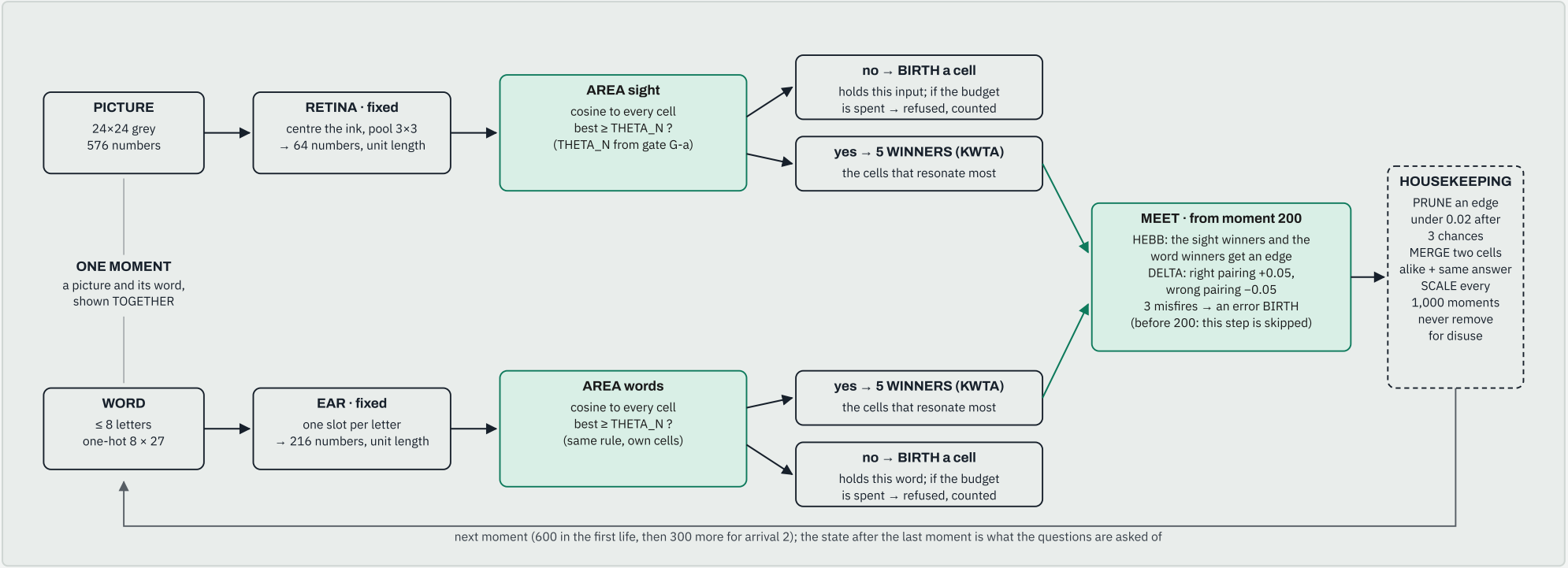


Figure 2 — Inside one moment. Green boxes are where the organism changes; the retina and the ear are fixed and never learn. The delta rule is the only place an edge shrinks for being wrong; pruning and merging are the only places anything is removed, and never for disuse.

5 · What waits on Ahmad

- **D-G1** — development only in rung 0, a lineage only in rung 2 on a substrate that has learned. Recommended yes.
- **D-G2** — a synthetic ten-thing world first, his own pictures and words in rung 1. Recommended yes.
- The one-sentence definition of "it works" above, if he wants it changed.

> Predictions written before any run (design file §7): (i)–(iii) and (v) likely PASS; (iv)'s 5 % pruning bar may FAIL, read then as a finding about the rule; (vi) is the uncertain one, a false-abstention rate above 0.20 the likely failure; (vii) PASS by construction; (ix) unknown.